

3. You are providing anaesthetic care for a 65 year old male, who weighs 98 kg, with a height of 170 cm. He has hypertension and is a heavy smoker (20 pack-years).

(a) Outline the physiological response to the establishment of pneumoperitoneum and subsequent steep Trendelenberg positioning (50%).

(b) You observe a steady increase in his end-tidal carbon dioxide tension, from 35 to 70 mmHg, during the first hour of anaesthesia with an associated increase in heart rate to 110 bpm. Describe your assessment (50%) and management (50%).

(Modified from FHKCA 2018)

Examiner Report

10/16
passed

Q2. This question asked for a detailed outline of the physiological responses to pneumoperitoneum and then steep Trendelenburg position. This was the best answered question in scenario A. Respiratory changes were well described. Some cardiovascular changes were confused or incorrectly sequenced. Half of the candidates described changes in renal physiology and cerebral circulation.

3/16
passed

Q3. This question asked for a detailed response in relation to a rising end-tidal carbon dioxide levels with tachycardia. The candidates needed to generate a reasonable differential diagnosis and then described their approach. Candidates who did poorly either had little in the way of differential diagnosis or decided to manage the situation with a crisis algorithm involving cancelling surgery rather than a more reasonable approach discussing likely causes and interacting appropriately with their surgical colleagues.

Sample Answer

(a) Physiological responses arising from Establishment of Pneumoperitoneum

CVS Higher IAP leading

Initial autotransfusion (mechanical displacement) from splanchnic circulation

Followed by impeded venous return, and drop in preload

Increase in systemic vascular resistance (afterload)

Vagal stimulation leading to bradycardia

Hypercarbia leading to sympathetic stimulation with tachycardia/ arrhythmias and increased contractility, but may cause direct myocardial depression at higher PaCO₂ > 65mmHg

Right shift of OHDC due to respiratory acidosis, increased O₂ release at tissue level

Respi Diaphragmatic splinting, with increased airway pressures, risk of barotrauma

Reduced FRC with shorter apnoeic time

Hypercarbia (secondary to systemic absorption of CO₂) leading to

- Increased pulmonary vascular resistance

- Increased respiratory drive, patient-ventilator dyssynchrony if not well paralysed

Worsens atelectasis, V/Q mismatch

Renal Reduced renal perfusion, leading to lower urine output

GI Impaired splanchnic circulation, leading to bowel oedema and post-op ileus

Increased aspiration risk

CNS Raised ICP from cerebral vasodilation secondary to hypercarbia, impeded CSF venous drainage

Others Venous stasis (lower limbs) from compression of veins, potentially increasing DVT risk

Steep Trendelenburg Positioning

CVS Improve venous return and preload

Respi Further diaphragmatic splinting, with increased airway pressures.

Further reducing FRC with shorter apnoeic time

Further worsening atelectasis and V/Q mismatch

Risk of one lung ventilation if inadvertent endobronchial intubation occurs

GI Further increasing aspiration risk

CNS Further worsening ICP

Others Facial/ upper airway oedema secondary to venous congestion.

Ocular perfusion pressure may also be reduced in prolonged steep Trendelenburg, risk peri-op visual loss from ischemia/ optic nerve injury.

(b)

With a steady increase in ETCO₂ from 35 to 70mmHg with associated increased in HR to 110 BPM, I am concerned about:

- Hypoventilation from malpositioned ETT/ pneumoperitoneum/ steep Trendelenburg position, which might be made worse by patient's obesity
- Inadequate analgesia/ anaesthetic depth/ paralysis with patient-ventilator desynchrony
- Pneumothorax, which could be contributed by patient's underlying COPD (emphysematous lungs) in view of his heavy smoking history.
- Bronchospasm, which again could be contributed by possible underlying COPD
- Subcutaneous emphysema (surgical cause)
- Equipment failure (rebreathing, exhausted soda lime)
- Reflex tachycardia from hypovolaemia from surgical bleed
- Conditions associated with hypermetabolic state (malignant hyperthermia, thyroid storm, sepsis, pheochromocytoma)

I would inform surgeon to stop surgery. Deflate pneumoperitoneum. Level bed. Followed by simultaneously assess and resuscitate patient according to ABC principles

Airway	Confirm <u>correct ETT placement</u> (check ETT depth marking at lips, no kinks in circuit/ETT) Check for <u>high airway pressures/ obstructive pattern</u> on capnography Manual bag-mask ventilation to check for <u>blocked ETT/ poor lung compliance/ chest wall rigidity</u> <u>Suction ETT</u> if suspected secretions/ mucous plugging.
Breathing	Titrate <u>FiO₂/PEEP</u> to maintain <u>SpO₂ >95%</u> Auscultate bilateral lungs for equal breath sounds, to exclude <u>bronchospasm, pneumothorax, endobronchial intubation</u> Check that <u>soda lime</u> is not exhausted, <u>insp/exp valves</u> functioning properly Check <u>ABG</u> to correlate ETCO ₂ with PaCO ₂
Circulation	Check and ensure that rest of vital signs are normal. In particular, for hypotension (BP), arrhythmia/ ischaemia (ECG), hyperthermia (temp), and hypoxia (SpO ₂) Optimise <u>anaesthesia depth, analgesia, and fluid resuscitation</u> to correct tachycardia

I would see if ventilation improves, with resolution of hypercarbia and tachycardia.

- Once tachycardia and hypercarbia resolves, to restart surgery with gradual and gentle pneumoperitoneum and less steep Trendelenburg position
- If patient is still unable to tolerate pneumoperitoneum/steep Trendelenburg, to consider converting to open surgery (with optimisation of analgesia).
- Patient will need to be monitored for extended period in recovery room, to consider sending to High Dependency if there are any respiratory/ventilation concerns.

If there is persistent hypercarbia/ tachycardia/ hyperthermia/ muscle rigidity, to declare crisis situation and manage patient as per Malignant Hyperthermia pathway.